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# Linux Filesystem

## 1. What is FileSystem?

A file system is a method used by the operating system to organize, store, and manage data on a storage device like a hard disk. It defines how data is stored, accessed, and retrieved

## 2. Windows vs Linux

Windows path: C:\Users\<user_name>\Downloads

Linux: /home/<user_name>/<file_name>

Windows is case insensitive and Linux is case sensitive

## 3. Types of File Systems (Common)

ext4 → most common in Linux
xfs → used in enterprise systems
ntfs → Windows
fat32 → USB drives

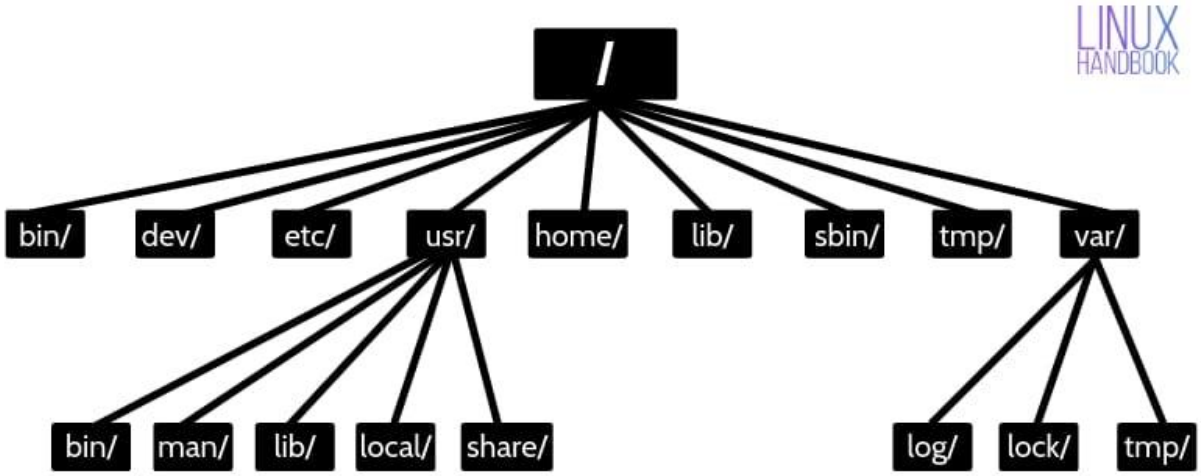
## 4. Difference between ext4 and XFS

ext4 and XFS are both Linux file systems. ext4 is more general-purpose and widely used, while XFS is designed for high performance and large-scale systems, especially for handling large files and parallel operations.

## Key Differences

| Feature | ext4 | xfs |
|---|---|---|
| Type | General purpose | High-performance |
| Performance | Good | Better for large data |
| File Size Support | Up to ~16TB | Very large (petabytes) |
| Resize | Can shrink & grow | Only grow (cannot shrink) |
| Speed | Moderate | Faster for large files |
| Journaling | Yes | Yes (more efficient) |
| Use Case | Default Linux FS | Enterprise / Big data |

## 5. Linux File System Structure




**/**

**Root directory** — the top of the filesystem; everything starts from here.

**/bin**

Essential user commands (like ls, cp, mv) required for basic system operation.

**/sbin**

System/admin commands used for system maintenance (mainly for root user).

**/etc**

Stores system-wide configuration files (users, services, network configs).

**/home**

Contains personal directories for all regular users.

**/root**

Home directory of the root (admin) user.

**/var**

Stores variable data like logs, cache, mail, and application data.

**/tmp**

Temporary files; usually cleared automatically after reboot.

**/usr**

Contains user-installed applications, libraries, and documentation.

**/opt**

Used for installing third-party or optional software.

**/dev**

Contains device files representing hardware (disks, USB, etc.).

**/proc**

Virtual filesystem showing process and system information (CPU, memory, running processes).

**/boot**

Contains bootloader files and Linux kernel required for system startup.

**/mnt**

Temporary mount point for manually mounted filesystems.

**/media**

Auto-mounted removable media like USB drives and CDs.

**/lib & /lib64**

Shared libraries required by system programs and binaries.

# Users & Groups

## 1. What is a User?

A **user** is a person or system account that can log into Linux and perform actions.

| **Example:** | **Login Command** |
|---|---|
| • Root -> admin user | sudo su |
| • Steve -> normal user | su - <user_name> |

## 2. What is a Group?

A **group** is a collection of users.

**Uses:**

- To manage permission easily.
- Instead of giving access to each user, you give access to a group.

**Example:**

- Group: Developers
- Users: ram, shyam , alex

**Commands:**

```
useradd john      # create user
adduser john      # advanced mode
groupadd dev      # create group
userdel john      #delete user
groupdel dev      #delete group
usermod -aG dev john   # add user to group
gpasswd –d john dev  #remove john from dev group
```

id <user_name>

output: uid=1001(user_name) gid=1004(group_name) groups=1004(group_name),1001(dev)

**User ID** | **Group ID**

# File Permissions

## File Permissions (rwx)

Every file/directory has permissions that define:  Who can read, write, or execute it

**Three Types of Permissions:**

| Symbol | Meaning | Example |
|---|---|---|
| r | Read | View file |
| w | Write | Edit file |
| x | Execute | Run file |

**Three Levels of Access:**

| Level | Meaning |
|---|---|
| User (u) | Owner of file |
| Group (g) | Group members |
| Others (o) | Everyone else |

**Each permission has a number:**

| Permission | Value |
|---|---|
| r (read) | 4 |
| w (write) | 2 |
| x (execute) | 1 |

**Example1:**

-rwxr-xr--

| **Section** | **Meaning** |
|---|---|
| Rwx | Owner → full access |
| r-x | Group → read & execute |
| r-- | Others → read only |

**Easy Trick:** Always read in groups of 3:

**rwx | r-x | r--**

**u g o**

**<u>Change Persmission:</u>**

chmod 755 file.txt

7 → rwx

5 → r-x

5 → r-x

**<u>Example2:</u>**

mkdir /tmp/shared

chmod 1777 /tmp/shared

1 777

| └── Normal permissions (rwx for all)

└─── Special permission (sticky bit)

Owner → full access

Group → full access

Others → full access

So everyone can read, write, create files

**What is that 1? (Sticky Bit)**

1 = Sticky Bit

**What Sticky Bit does**

**Rule:**

Users can create files, but can only delete their own files

**<u>Example3:</u>**

chmod g+s /tmp/project

This command sets the SGID (Set Group ID) on the /tmp/project directory.

## What SGID does?

**When SGID is applied on a directory:**

All new files and folders created inside will inherit the same group as the directory.

**Without SGID**

/project → group = dev

User john creates file:

file.txt → group = john

Problem:

Group mismatch

Team cannot access properly

**With SGID (After chmod g+s)**

/project → group = dev

User john creates file:

file.txt → group = dev

Solution:

All team members (dev group) can access

# Ownership

**Every file has:**

- Owner (user)
- Group

**Example1:**

-rw-r--r-- 1 john dev file.txt

- Owner = john
- Group = dev

**Change Ownership**

- chown john file.txt **# change owner**
- chown john:dev file.txt **# change owner + group**

**Example2:**

-rwxr----- 1 john dev script.sh

- Owner (john) → full access
- Group (dev) → read only
- Others → no access

**Example3:**

- chown john:dev file.txt
- chmod 770 file.txt
- Owner + group → full access
- Others → no access

Users = who is using system

Groups = collection of users

Permissions = what they can do

Ownership = who controls the file

**#Recursive flag:** It inherits the command property.

Ex: file path: /home/ubuntu/file.txt

chmod 755 /home/ubuntu  #it will change permission for only ubuntu dir

chmod  -R 755 /home/ubuntu  #it will change permission for ubuntu along with all files & dirs inside ubuntu.

**-R → recursive (safe meaning across commands, applicable everywher)**

**-r → depends on command (cannot use in chmod, chown)**

# Process Lifecycle

## How a process lives & dies

### 1. What is a Process?

A **process** is equal to a running program

Example:
sleep 100  #command
This creates a process

### 2. Lifecycle stages

A process goes through different states:

### New → Ready → Running → Waiting → Terminated

**a. New (Created)**

Process is created

Example: you type a command

**b. Ready**

Waiting for CPU

In queue

**c. Running**

CPU is executing it

**d. Waiting (Sleeping)**

Waiting for I/O (disk, network, etc.)

Example:

sleep 10 #command

It is in **waiting state**

**e. Terminated**

Process finished or killed

**How to check process states**

ps aux #command

Look at column:

- R → Running
- S→ Sleeping
- Z → Zombie (important )

## 3. What is Zombie process?

-> Process is dead but still in table

-> Parent didn’t read its status

**Kill a process**

kill <PID>  #command

kill -9 <PID>  # force kill command

# Foreground vs Background Process

## 1. Foreground process

-> Runs in terminal and blocks it

Example:

sleep 20 #command

You cannot type anything until it finishes

## 2. Background process

-> Runs in background → terminal is free

Example:

sleep 100 &  #command

Output:

[1] 1234

-> 1 → job id

-> 1234 → PID

### Move running process to background

a. Run:

sleep 100 #command

b. Press:

Ctrl + Z

Suspends process

c. Then:

bg #command

Runs in background

### Bring back to foreground

fg #command

**See background jobs**

jobs #command

**Summary**

| Type | Behavior |
|---|---|
| Foreground | Blocks terminal |
| Background | Runs freely |

# System Monitoring (Very Important )

## 1. Why monitoring?

**To check:**

-> CPU usage

-> Memory usage

-> Running processes

-> System load

a. **Check CPU usage**

**i. top** #command

Shows:

-> CPU %
-> Memory
-> Running processes

**ii. Better version**

**htop** #command

More user-friendly (install if not present)

b. **Check memory**

free -h #command

**Output:**

-> total

-> used

-> free

c. **Check processes**

ps aux #command

**d. Check specific process**

ps aux | grep nginx #command

**e. System load**

uptime #command

Shows load average

**f. Disk usage**

df –h #command

**g. Real-time process tracking**

top –p <PID> #command

# # htop Command explanation:

## 1. CPU Colors Explanation

| Color | Meaning | Good/Bad |
|---|---|---|
| Green | User processes | ✅ Normal |
| Red | Kernel/system | ⚠️ Watch |
| Blue | Nice processes | ✅ Fine |
| Yellow | I/O wait | ❌ Disk issue |
| Purple | Steal time | ❌ VM issue |
| Grey | Idle | ✅ Good |

**Case 1: Mostly Green**

-> Normal application load
✓ Healthy system

**Case 2: High Red**

**Kernel busy**
Possible causes:

-> Heavy I/O
-> Network issues
-> Driver problem

**Case 3: High Yellow**

**Disk is slow**
Common in:
-> Database servers
-> Logging-heavy systems

**Case 4: High Purple**

**Cloud issue**

-> Shared CPU overloaded

## 2. Memory Colors Explained

**Green → Used Memory (Applications)**
Memory used by:
-> Running programs (Java, Python, etc.)
-> Services (nginx, mysql)
This **is real usage**

**Blue → Buffers**
Memory used by kernel for:
-> Temporary data
-> Disk operations
✓ Small portion usually
✓ Helps system performance

**Yellow → Cache**

Memory used for:

-> File caching

-> Frequently accessed data

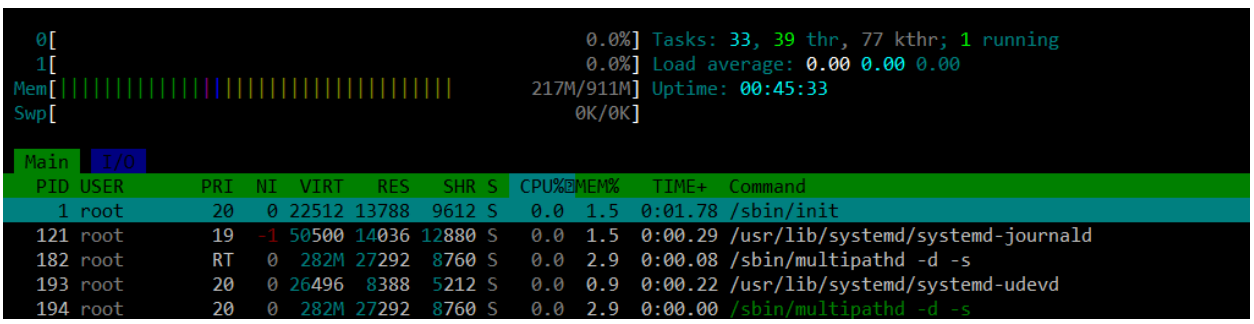
✓ VERY IMPORTANT:

This memory can be freed anytime if needed

Grey → **Free Memory**

-> Completely unused RAM

| Color | Meaning | Important? |
|---|---|---|
| Green | App memory | ✅ Yes |
| Blue | Buffers | Minor |
| Yellow | Cache | ✅ Very important |
| Grey | Free | Not critical |




**a. PID:** Process ID

**b. User:** Name of users

**c. PRI(Priority):** Lower number -> Higher priority, kernel decides this based on NI

**d. NI (Nice value):** controls priority manually. Ex: -20(highest priority) , 19(Lowest Priority), 0 (Default Priority)

**e. VIRT(Virtual Memory):** Total memory used by process including everything (RAM, SWAP, Shared lib, reserved memory). *Note: VIRT is often large and misleading.*

**f. RES (Resident memory):** Actual RAM used right now.

**g. SHR (Shared Memory):** Memory shared with other processes. Ex: shared libraries, common resources.

**h. S (State):** Process state.

## # top command explanation:

```
top - 11:48:58 up  1:02,  3 users,  load average: 0.00, 0.00, 0.00
Tasks: 116 total,   1 running, 115 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.2 us,  0.3 sy,  0.0 ni, 99.5 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :    911.5 total,    271.6 free,    382.3 used,    416.1 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    529.2 avail Mem
```

i) **11:48:58** -> current system time

ii) **up 1:02min** -> system running since last reboot

iii) **3 users** -> 2 logged-in users

iv) **load average** -> system load.

0.00, 0.00, 0.00 -> Last 1 min, 5 min, 15 min resp.

Ex: 0.00 -> no load (idle system), 1.00 -> CPU fully used, > CPU cores -> overloaded.

v) **Tasks:**

```
Tasks: 116 total,   1 running, 115 sleeping,   0 stopped,   0 zombie
```

| Field | Meaning |
|---|---|
| total | total processes |
| running | currently using CPU |
| sleeping | waiting (normal) |
| stopped | paused (Ctrl+Z) |
| zombie | dead but not cleaned |

Currently 1 running -> very light load.

115 sleeping -> normal

0 zombie -> healthy system

vi) **CPU usage:**

```
%Cpu(s):  0.2 us,  0.3 sy,  0.0 ni, 99.5 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
```

current -> 99.5 id  CPU is 99.5% idlle

| Field | Meaning |
|---|---|
| us | User processes |
| sy | System (kernel) |
| ni | Nice (low priority) |
| id | Idle |
| wa | Waiting for I/O |
| hi | Hardware interrupts |
| si | Software interrupts |
| st | Stolen (VM usage) |

vii) **Memory (RAM):**

```
MiB Mem :    911.5 total,    271.6 free,    382.3 used,    416.1 buff/cache
```

buff/cache is not wasted. Linux is using free RAM as cache to improve performance.

| Field | Meaning |
|---|---|
| total | total RAM |
| free | unused RAM |
| used | actively used |
| buff/cache | cached memory |

viii) **Swap:**

```
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    529.2 avail Mem
```

disk used as RAM (backup memory.)
current -> No swap configured, available memory for swap: 592.2 MB

# Inode, Kernel, and Daemon

## Inodes

Inode is a data structure in Linux that stores metadata about a file such as permissions, owner, size, and disk location. It does not store the file name or actual file content.

Key Points:
- Each file has a unique inode number
- Stores metadata like size, permissions, owner
- Does not store file name or content
- Used by OS to locate file data

## Inode Number

An inode number is a unique identifier assigned to every file in a Linux filesystem.
It is used by the system to locate the file's metadata and actual data on disk.
Ex: 123456 file.txt -> inode_number file_name

**Commands:**
ls -i file.txt
df -i
stat file.txt

Important Concept:
Disk can be full due to space (df -h) or inode exhaustion (df -i)

## Kernel

The kernel is the core component of the Linux OS that interacts directly with hardware and manages system resources like CPU, memory, and devices. It acts as a bridge between applications and hardware.

Key Functions:
- CPU management
- Memory management
- Device handling
- System calls execution

Working Flow:
Application → Kernel → Hardware

Example:
When a command like 'ls' is executed, the kernel interacts with the disk and returns the file list.

## Daemon

A daemon is a background process that runs continuously to provide system services or perform tasks. It usually starts at boot and runs without user interaction.

Examples:
sshd → SSH service
cron → scheduled tasks
httpd/nginx → web server

Commands:
ps -ef | grep sshd
systemctl status ssh

## Kernel vs Daemon

Kernel:
- Core of OS
- Manages hardware
- Runs in kernel space

Daemon:
- Background service
- Provides system/application services
- Runs in user space

## Summary

Inodes store file metadata, kernel manages hardware/resources, and daemons provide background services.

# Disk Structure

## Definition

Disk structure refers to how a storage device is organized into partitions and file systems so the operating system can store and manage data efficiently. It defines how raw disk space is divided and used.

## Basic Idea

A disk (like HDD/SSD) is not used directly. It is divided into partitions, and each partition is formatted with a file system.

## Disk Structure Flow

Disk → Partition → File System → Mount → Access Files

## Important Components

1. Disk
Physical storage device
Example: /dev/sda, /dev/sdb

2. Partition
Logical division of disk
Example: /dev/sda1, /dev/sdb1

3. File System
Organizes data
Example: ext4, xfs

4. Mount Point
Location where disk is attached
Example: /mnt, /data

## Example

Suppose you have a new disk /dev/sdb
1. Create partition → /dev/sdb1
2. Format it → ext4 or xfs
3. Mount it → /data
4. Store files

## Visual Understanding

```
/dev/sdb (Disk)
  └── /dev/sdb1 (Partition)
      └── ext4 (File System)
          └── \data (Mount Point)
                └──Files & Data
```

## Practical Commands

Check disks:
lsblk

Create partition:
fdisk /dev/sdb

Format disk:
mkfs.ext4 /dev/sdb1

Mount disk:
mount /dev/sdb1 /data

## Real-Life Analogy

Disk = Land
Partition = Plots
File system = Layout plan
Files = Houses

## Key Understanding

- Disk must be partitioned before use
- Partition must be formatted
- Then it can store data

## Summary

Disk structure is the organization of a disk into partitions and file systems so data can be stored and accessed properly.

# Mounting

## Definition

Mounting is the process of attaching a filesystem (disk or partition) to a specific directory so that it becomes accessible to the system. Without mounting, the OS cannot access the data on that disk.

## Basic Idea

In Linux, storage is not used directly. It must be mounted to a directory (mount point) before accessing files.

## Mounting Flow

Disk → Partition → File System → Mount → Access Files

## Example

/dev/sdb1 is mounted to /data.
After mounting, /data shows the contents of /dev/sdb1.

## Visual Structure

/dev/sdb1 (disk partition)
↓
mounted on
↓
/data (directory)

## Temporary vs Permanent Mount

Temporary Mount:
mount /dev/sdb1 /data (lost after reboot)

Permanent Mount:
Edit /etc/fstab and add:
/dev/sdb1 /data ext4 defaults 0 0

## Important Commands

Check mounted disks:
mount
lsblk

## Unmounting

umount /data
or
umount /dev/sdb1

## Important Points

Directory must exist before mounting.
Cannot unmount if the disk is in use.
Always verify after mounting.

## Scenario

If disk is attached but not visible:
1. lsblk
2. mount
3. Check /etc/fstab

## Summary

Mounting connects a disk or partition to a directory so the system can access its data. Without mounting, storage devices remain unusable.

# Practical: AWS EBS Volume Setup Guide (Linux)

**Step 1:** Verify Attached Volume
Run: **lsblk**
You will see new disk like:
**nvme1n1 10G disk**

**Step 2:** Create Partition
Run: **sudo fdisk /dev/nvme1n1**

**Inside fdisk:**
n → new partition
p → primary
1 → partition number
Enter → default start
Enter → default end
w → save

**Step 3:** Verify Partition: **lsblk**
Output:
**nvme1n1**
└**nvme1n1p1**

**Step 4:** Format Filesystem: **sudo mkfs.ext4 /dev/nvme1n1p1**

**Step 5:** Create Mount Point: **sudo mkdir /data**

**Step 6:** Mount Volume: **sudo mount /dev/nvme1n1p1 /data**

**Step 7:** Verify Mount: **df -h**

**Step 8:** Make Mount Permanent
Edit **fstab:**
**sudo nano /etc/fstab**
Add line: **/dev/nvme1n1p1 /data ext4 defaults,nofail 0 2**

Test: **sudo mount -a**

**Step 9:** Store Data
**cd /data**
**echo "Hello AWS Volume" > file.txt**
**<u>Explanation of fstab Entry</u>**

Line: **/dev/nvme1n1p1 /data ext4 defaults,nofail 0 2**

**Structure:**
**<device> <mount_point> <filesystem> <options> <dump> <fsck>**

**1. Device**
/dev/nvme1n1p1
Disk partition
nvme1n1 = disk
p1 = partition 1

**2. Mount Point**
/data
Location where disk is mounted
Access using: cd /data

**3. Filesystem**
ext4
Type of filesystem
Common in Linux

**4. Options**
defaults,nofail
defaults includes:
rw (read/write), suid, dev, exec, auto, nouser, async

**nofail:** Prevents boot failure if disk is missing

**5. Dump**
0: Backup option (usually not used)

**6. FSCK Order**
2
Filesystem check order
1 = root (/)
2 = other filesystems

**Practical2: Resize Disk**

**Step1:** increase volume in AWS
10GB → 20GB

**Step2:** verify new size
lsblk
You'll see disk increased but partition same.

**Step3:** Extend partition
sudo growpart /dev/nvme1n1 1

**Step4:** resize filesystem
For ext4:
sudo resize2fs /dev/nvme1n1p1

**Done**
Check: **df –h**

**Unmount command: sudo umount <mountpoint>**

# LVM (Logical Volume Manager) Basics

## Definition

LVM (Logical Volume Manager) is a storage management system that allows flexible allocation and resizing of disk space. It helps combine multiple disks and manage them as a single logical storage.

## Why LVM is Used

Normal partitions have fixed size and are hard to resize.
LVM allows easy resizing, combining multiple disks, and better storage management.

## LVM Structure

Disk → PV → VG → LV → Filesystem → Mount

## Components of LVM

**PV (Physical Volume):** A physical disk or partition used in LVM (e.g., /dev/sdb).

**VG (Volume Group):** A pool of storage created by combining multiple PVs (e.g., vg_data).

**LV (Logical Volume):** A virtual partition created from VG, used like a normal disk (e.g., lv_data).

## Visual Structure

```
/dev/sdb + /dev/sdc → VG (vg_data)
                ↓
            LV (lv_data)
                ↓
              /data
```

## Useful Commands

pvs → Check physical volumes
vgs → Check volume groups
lvs → Check logical volumes

## Advantages

Flexible storage management
Easy resizing
Combine multiple disks
Suitable for production environments

## Important Points

LVM sits between disk and filesystem.
Logical volumes behave like normal partitions.
Commonly used in servers and cloud environments.

## Summary

LVM allows flexible disk management by combining physical storage into logical volumes that can be resized easily.

## Practical Steps

**1. Create Physical Volume:** sudo pvcreate /dev/nvme1n1, **check:** sudo pvs

**2. Create Volume Group:** sudo vgcreate vg_data /dev/ nvme1n1, **check:** sudo vgs

**3. Create Logical Volume:** lvcreate -l 100%FREE -n lv_data vg_data, **check:** sudo lvs

**4. Format:** mkfs.ext4 /dev/vg_data/lv_data

**5. Mount:** mkdir /data -> mount /dev/vg_data/lv_data /data, **check:** df –h

lvcreate -l 100%FREE -n lv_data vg_data

**-L: Fixed size, Ex: 2G, 500M**

**-l: Relative size, Ex: 100%FREE, 50%FREE**

## Extend Logical Volume

**1. Resize Physical Volume (PV):** sudo pvresize /dev/nvme1n1

**2. Now Extend Logical Volume (LV):** sudo lvextend -l +100%FREE /dev/vg_data/lv_data

**3. Resize Filesystem:** sudo resize2fs /dev/vg_data/lv_data **(FOR xfs:** sudo xfs_growfs /data**)**

# IP Address

## Definition

An IP address (Internet Protocol address) is a **unique numerical identifier** assigned to a device on a **network**. It is used to **identify** and **communicate** between devices over a network.

## 1. Why IP Address is Needed

To identify devices
To send and receive data
To enable communication between systems

## 2. Simple Understanding

Think:

IP address = house address
Internet = city
Without address → data cannot reach the correct device

## 3. Example

192.168.1.10 (This is a typical IP address)

## 4. Types of IP Address

### 1. IPv4 (Most Common)

An IPv4 address is made up of 32 bits, divided into 4 octets.
Each octet contains 8 bits and is separated by dots.

**Octet:** An octet is a group of 8 bits in an IP address.

#### Structure of IPv4

192.168.1.10

**Breakdown:**

192 .168 .1 .10

Octet1  Octet2  Octet3  Octet4

### Range of Each Octet

**Each octet can have values:** 0 to 255

**Because:** 2^8 = 256 values (0–255)

## 2. IPv6

IPv6 (Internet Protocol version 6) is the newer version of IP addressing designed to replace IPv4.
It uses 128 bits, allowing a much larger number of unique IP addresses.

### Structure of IPv6

IPv6 is made of 8 groups (blocks).
Each group = 16 bits

**Example:** 2001:0db8:85a3:0000:0000:8a2e:0370:7334

**Breakdown:**

```
2001 : 0db8 : 85a3 : 0000 : 0000 : 8a2e : 0370 : 7334
  |      |      |      |      |      |      |      |
 Block1 Block2 Block3 Block4 Block5 Block6 Block7 Block8
```

### How It Is Written

Hexadecimal numbers (0–9, a–f)
Colon : separator

### Address Capacity

**IPv4:** 32 bits → ~4.3 billion IPs
**IPv6:**128 bits → extremely large (almost unlimited)

# 5. Public vs Private IP

## Private IP

Used inside local network
Example: 192.168.x.x, 10.x.x.x, 172.16.x.x

## Public IP

Used on internet. Provided by ISP

## 6. Static vs Dynamic IP

### Static IP

Fixed address. Does not change
Use: Servers

### Dynamic IP

Changes automatically. Assigned by DHCP
Use: Client devices (Desktop, Laptop, Mobile etc.)

## 7. Check IP in Linux

**ip a** or **hostname -I**

## 8. Real Example

- When you open a website:

Your system sends request from your IP and Server responds to your IP

## 9. Important Points

Every device must have an IP
No two devices can have same IP in same network
Used in all networking

## Final Summary

An IP address is a unique number used to identify devices and enable communication in a network.

# IP Classes

IP classes are a way to divide IPv4 addresses into different categories based on size and usage of networks.
They help in organizing IP addresses for small, medium, and large networks.

## 1. Why IP Classes Exist

To manage IP address allocation
To support different network sizes
To separate network and host portions

## 2. Main IP Classes

**There are 5 classes:** i) Class A  ii) Class B  iii) Class C  iv) Class D  v) Class E

## 3. Class A

**Range:** 1.0.0.0 – 126.255.255.255
**Default Subnet:** 255.0.0.0
**Use:** Very large networks
**Example:** 10.0.0.1

## 4. Class B

**Range:** 128.0.0.0 – 191.255.255.255
**Default Subnet:** 255.255.0.0
**Use:** Medium-sized networks
**Example:** 172.16.0.1

## 5. Class C

**Range:** 192.0.0.0 – 223.255.255.255
**Default Subnet:** 255.255.255.0
**Use:** Small networks
**Example:** 192.168.1.1

## 6. Class D

**Range:** 224.0.0.0 – 239.255.255.255
**Use:** Multicast (group communication)

## 7. Class E

**Range:** 240.0.0.0 – 255.255.255.255

**Use:** Experimental / research

## 8. Quick Trick to Identify Class

Look at **first octet**

| First Number | Class |
|---|---|
| 1–126 | A |
| 128–191 | B |
| 192–223 | C |
| 224–239 | D |
| 240–255 | E |

## 9. Easy Memory Trick

A → Big network

B → Medium

C → Small

## Final Summary

IP classes divide IPv4 addresses into groups (A, B, C, D, E) based on network size and purpose.

# What about 127.x.x.x ?

## Definition

The IP range 127.0.0.0 – 127.255.255.255 is reserved for loopback addresses.
It is used by a system to communicate with itself and is not used for normal networking.

### 1. Why 127 is Missing in Class A

Earlier we said **Class A: 1 – 126.** Because **127 is reserved.** So, it is excluded from usable **Class A range**

### 2. Most Common Loopback IP

**127.0.0.1**. It is Also called **localhost**

### 3. What is Loopback?

It means Your system talks to **itself**

**Example:** ping 127.0.0.1

**Result:**

Tests local network stack. Does NOT go to internet

### 4.Usage

Testing network services
Checking if TCP/IP is working
Local development (apps running on same machine)

### Final Summary

127.x.x.x is reserved for loopback (localhost) and is not used as a normal Class A network range.

## IP Address Count in Each Class

### Formula

Total IPs = 2^(number of host bits)

**Example:**

**Class A**

**Default subnet:** 255.0.0.0

**Network bits:** 8

**Host bits:** 24

**Total IPs:** 2^24 = 16,777,216

**Usable IPs:** 16,777,214 (Subtract network IP + broadcast IP)

***Note: Class D*** *has no usable* ***host IP*** *concept. It is only used for multicast. And* ***Class E*** *is only reserved for research.*

## Network Address

A network address is the first IP address in a network, used to identify the entire network.
It cannot be assigned to any device.

## Broadcast Address

A broadcast address is a special IP address used to send data to all devices in a network at once.
It is the last IP address of a subnet and cannot be assigned to any device.

# Subnet

A subnet is a smaller division of a network created by splitting a larger network into multiple smaller networks.
It helps in better management, security, and efficient use of IP addresses.

## Why Subnetting is Needed

- Reduce network congestion
- Improve performance
- Increase security
- Better IP management

**Example:** 192.168.1.0/24 | **Total IPs:** 256

## Subnet Mask

Subnet mask defines which part of IP is network and which is host.

**Example:** 255.255.255.0

## CIDR Notation

CIDR (Classless Inter-Domain Routing) is a shorthand way to represent the subnet mask using a number.

**Example:** /24.

## VLSM (Variable Length Subnet Mask)

VLSM is a subnetting technique of creating different sizes of subnets.

## Why VLSM is Needed

Avoid IP wastage
Different departments need different IP sizes
More efficient than fixed subnetting

**Example:** 192.168.1.0/24, 192.168.1.0/26, 192.168.1.0/27

## Subnet count Formula:

**(2^[Total_Octet_Size] - [CIDR Notation])**

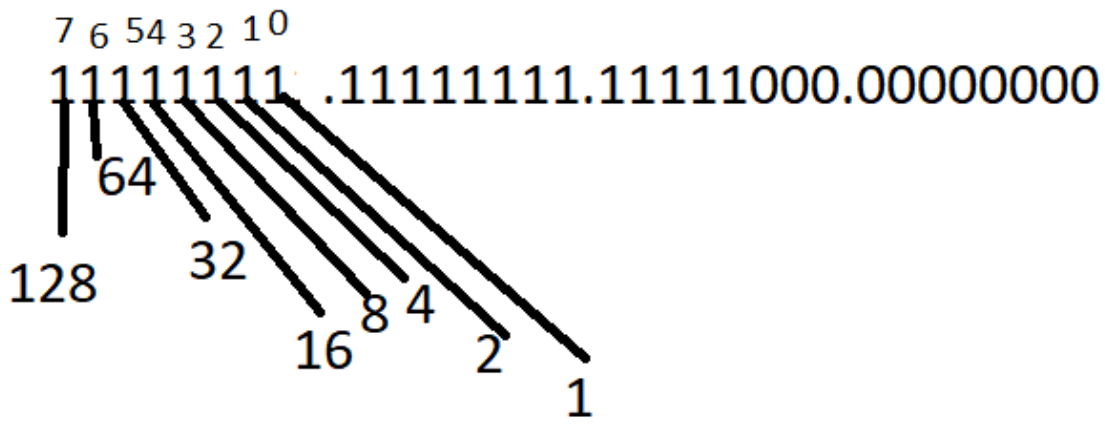
**Example:**
**How many IPs in this subnet range 192.168.1.0/21?**

**Step1: 32 – 26 = 6**
**Step2: 2^6 = 64**

**Number of IPs: 64**
**Number of usable IP: 64 – 2 = 62**

## Subnet mask count Formula:

**128+64+32+16+8+4+2+1 = 255 – size of 1 octet**




**Example:**
**255-7 = 248| 255.255.248.0**

# DNS(Domain Name System)

## Definition

DNS (Domain Name System) is a system that converts human-readable domain names into IP addresses. It allows users to access websites using names instead of remembering IP addresses

## 1. Why DNS is Needed

**Humans remember:** google.com
Machines understand: **142.250.183.14**

**DNS converts:** Name → IP

## 2. Simple Understanding

**DNS =** Phonebook of Internet
**Domain =** Name
**IP =** Phone number

## 3. Types of DNS Servers

### i) Recursive Server (DNS Resolver)

A DNS resolver is a server that receives a domain name request and finds the corresponding IP address by querying other DNS servers.
It acts as the middleman between the user and the DNS system.

## Where Resolver Exists

ISP DNS (Airtel, Jio etc.)
Public DNS: 8.8.8.8(google), 1.1.1.1(Cloudflare)

Example:
**Search:** google.com
**Resolver =** helper who finds the IP for you

### ii). Root Server (Global root servers)

Global root servers are the top-level DNS servers that handle the first step of DNS resolution and direct queries to the correct TLD servers.

There are 13 logical root server systems, distributed worldwide for reliability and performance.

### a). Why Only 13 Root Servers?

- Historically, DNS used limited packet size (512 bytes). So, Only 13 root server addresses could fit
- Important, These are 13 logical servers, not just 13 machines.

### b). Names of Root Servers

They are labeled:
A.root-servers.net
B.root-servers.net
C.root-servers.net
...
M.root-servers.net
Total: A to M = 13 root servers

### c). Who Manages Them?

Different organizations manage them, for example: ICANN, Verisign, NASA

## iii). TLD Server

A TLD (Top-Level Domain) server is a DNS server that manages domain extensions like .com, .org, .in and directs queries to the correct authoritative DNS server.
It does not provide the final IP address, but points to where the domain is hosted.
Example: .com, .org, .in

## iV). Authoritative DNS

An authoritative DNS server is the DNS server that holds the actual DNS records of a domain and provides the final answer (IP address).
It is the source of truth for a domain..

## # Nameserver

A nameserver is a DNS server that responds to domain name queries and helps resolve domain names into IP addresses.
It stores or retrieves DNS records for a domain.

Example: ns1.example.com, ns2.example.com

## Types of Nameservers

**1. Recursive Nameserver (Resolver)**
Example: 8.8.8.8
Does lookup for you

**2. Authoritative Nameserver**
Example: ns1.google.com

Stores real DNS records

## 4. How DNS Works (Step-by-Step)

**Example:**
**You type:** www.google.com

**Step 1:** Check Local Cache
**System checks:**
- Browser cache
- OS cache

**Step 2: Query DNS Server**
**If not found:** Request goes to DNS resolver

**Step 3: DNS Resolution Process**
Root Server → TLD Server → Authoritative Server

**Step 4: Get IP**
www.google.com → 142.250.x.x

**Step 5: Connect to Server**
Browser connects using IP

## 5. DNS Records (Definition)

DNS records are entries in a DNS database that map domain names to IP addresses or define how a domain behaves.
They tell the DNS server how to respond to queries.

### i). A Record (Address Record)

An A (Address) record is a DNS record that maps a domain name to an IPv4 address.
It tells the system which IP address a domain should connect to.
**Example:** example.com → 192.168.1.10
Most commonly used

### ii). AAAA Record

An AAAA record is a DNS record that maps a domain name to an IPv6 address.
It is similar to an A record, but it works for IPv6 instead of IPv4.
**Example:** example.com → 2001:db8::1

### iii). CNAME Record (Canonical Name)

A CNAME (Canonical Name) record maps one domain name to another domain name.
It is used to create an alias instead of pointing directly to an IP address.
**Example:** www.example.com → example.com
Used for Aliases

### iv). MX Record (Mail Exchange)

An MX (Mail Exchange) record is a DNS record that specifies which mail server is responsible for receiving emails for a domain.
It directs email traffic to the correct mail server.
**Example:** example.com → mail.example.com
Used in email systems

### v). NS Record (Name Server)

An NS (Name Server) record specifies which DNS servers are authoritative for a domain.
It tells the internet where to go to find the DNS records of that domain.
**Example:** example.com → ns1.example.com

### vi). PTR Record (Pointer Record)

Maps IP address back to domain name (reverse DNS).
**Example:** 192.168.1.10 → example.com

### vii). TXT Record

Stores text information for domain verification and security.
**Example:** "v=spf1 include:_spf.google.com ~all"
Used for SPF, DKIM

### viii). SOA Record (Start of Authority)

Contains administrative information about the domain.
It Includes Primary nameserver, Email, Serial number

## 6. Check DNS in Linux

nslookup google.com or dig google.com

## 7. DNS in System

**File:** /etc/resolv.conf

**Example:**
nameserver 8.8.8.8

## 8. Real Example

**ping google.com**

DNS resolves → IP → ping works

### #Key Understanding

- DNS = Name → IP
- Works in background
- Essential for internet

## Final Summary

DNS converts domain names into IP addresses so users can access websites easily.

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# Gateway

## Definition

A gateway is the entry/exit point in a network that allows communication with other networks.
It is usually represented by an IP address configured on a device.

### 1. Simple Understanding

Your system → local network

Gateway → door to outside world

Without gateway, you can talk inside network. But cannot reach internet

**Example**

Your system IP: 192.168.1.10

Gateway: 192.168.1.1. This IP assigned in router only.

### 2. Why Gateway is Needed

- Connect different networks

- Send traffic outside local network

- Route packets to internet

## 3. How Gateway Works

**Step-by-Step**

1. You send request to external IP

2. System checks: “Is this in my network?”

**Case 1:** Same Network

192.168.1.20 → Direct communication

**Case 2:** Different Network

Request Send to gateway, then goes to server

**Visual Flow**

PC (192.168.1.10)
↓
Gateway (192.168.1.1)
↓
Internet
↓
Server

## 4. Routing Logic

Destination inside → send directly
Destination outside → send to gateway

## 5. Gateway Types

Gateways can be categorized based on their function and the type of traffic or networks they connect. Different gateways serve different purposes in a network.

### i. Default Gateway

The main gateway used to send traffic outside the local network when no specific route exists.

### What is a Specific Route?

A specific route is a routing rule that tells the system exactly how to reach a particular network using a specific gateway.

### Router

A router is a network device that forwards packets between different networks based on routing tables.

### Routing Table

A routing table is a list of rules used by a system or router to decide where to send network packets.
It maps destination networks to the next hop (gateway) and interface.

**Example**
**Your system IP:** 192.168.1.10

**Routing table:**
10.0.0.0/8 via 192.168.1.2 (If request destination is 10.0.0.0/8 then 192.168.1.2 gateway will use. )
default via 192.168.1.1

**# Check in Linux: ip route**

**Example:**

default via 192.168.1.1 dev eth0

192.168.1.1 = gateway

### ii. Network Gateway

A device that connects two different networks and routes traffic between them.

**Example:** Home router connecting LAN → Internet

### iii. Application Gateway (Proxy Gateway)

A gateway that operates at application level and manages traffic for specific applications.

**Example:** Web proxy server can Filter traffic Cache content.

### iv. NAT Gateway

A gateway that translates private IP addresses into public IP addresses for internet communication.

**Example:** Cloud NAT (AWS/Azure)**,** Home router NAT

### v. VPN Gateway

A gateway that allows secure communication between networks over the internet using encryption.

**Example:** Site-to-site VPN**,** Remote access VPN

### vi. Internet Gateway (Cloud Concept)

A gateway used in cloud environments to allow resources in a private network to access the internet.

**Example:** AWS Internet Gateway

## 7. Important Points

- Gateway must be in same network

- Usually router IP

- Only one default gateway

## 8. Common Issue

**If gateway is wrong:** Internet not working but local network works

**Final Summary**
A gateway is a router that connects your local network to other networks and acts as the exit point for external communication.

# Port

## Definition

A port is a logical communication endpoint used by a computer to identify a specific process or service on a system.
It allows multiple services to run on a single IP address.

### 1. Why Ports Are Needed

**One system runs multiple services:** Web server**,** SSH**,** Database

**Ports help identify:** Which service should receive the data

### 2. Port Number Range

| Range | Type |
|---|---|
| **0–1023** | **Well-known ports** |
| **1024–49151** | **Registered ports** |
| **49152–65535** | **Dynamic/ephemeral** |

### 3. Common Ports (Important)

| Port | Service |
|---|---|
| 22 | SSH |
| 80 | HTTP |
| 443 | HTTPS |
| 21 | FTP |
| 25 | SMTP |
| 53 | DNS |

### 4. Example

**When you open website:** google.com → port 443

**Flow:**

IP + Port = Destination
142.x.x.x:443

## 5. Check Open Ports (Linux)

netstat –tulnp or ss –tulnp

**Key Difference**
ss is faster, more efficient, and more detailed than netstat.

## Key Points

- Ports identify services
- Same IP can have many ports
- Required for communication

## Final Summary

Ports are logical endpoints that allow multiple services to run on a single system by identifying specific applications using port numbers.

# Protocol

## Definition

A protocol is a set of rules that define how data is transmitted and received over a network.
It ensures that devices can communicate correctly and understand each other.

# Types of Protocols

## 1 Application Layer Protocols

Used by users/apps

**Examples**

| Protocol | Purpose |
|---|---|
| HTTP | Web browsing |
| HTTPS | Secure web |
| FTP | File transfer |
| SMTP | Sending emails |
| DNS | Name → IP |

## 2 Transport Layer Protocols

Control how data is delivered

◆ **TCP (Transmission Control Protocol)**

- Reliable, connection-based communication

### Features
- Data guaranteed
- Ordered delivery
- Slower

◆ **UDP (User Datagram Protocol)**

Fast, connectionless communication

### Features
- No guarantee
- Faster
- Used in streaming

### 3 Network Layer Protocols

Handle addressing and routing

**Examples**

| **Protocol** | **Purpose** |
|---|---|
| IP | Addressing |
| ICMP | Error messages |
| ARP | IP → MAC |

### Protocol Stack Example

**When you open a website:**

Browser (HTTP)
↓
TCP (port 443)
↓
IP (destination address)
↓
Internet

### Key Characteristics

- Defines rules
- Ensures communication
- Works in layers

## Final Summary

Protocols are standardized rules that allow devices to communicate over a network, handling everything from data formatting to delivery.

# ICMP

ICMP (Internet Control Message Protocol) is a network layer protocol used for sending error messages and operational information between devices.
It helps in diagnosing network issues and checking connectivity.

**Simple Understanding**

**ICMP = network health checker**

**It tells:**

Is destination reachable?

Is there an error?

**Most Common Example**

ping command

ping google.com

**Uses ICMP**

**Sends:** Echo Request

**Receives:** Echo Reply

## ARP

ARP (Address Resolution Protocol) is used to find the MAC address of a device when its IP address is known.
It works within a local network to enable communication.

**Simple Understanding**

IP = logical address

MAC = physical address

**ARP = "IP → MAC converter"**

## Why ARP is Needed

When your system sends data:

It needs MAC address, not just IP

**Problem:** “I know IP, but how to find MAC?”

**Solution:** ARP

## MAC Address

A MAC (Media Access Control) address is a unique hardware identifier assigned to a network interface card (NIC).
It is used to identify devices within a local network.

IP address → logical (can change)

MAC address → physical (fixed)

### Example of MAC Address

**00:1A:2B:3C:4D:5E**

### Format:

6 pairs (hexadecimal)

48 bits total

### Structure

```
00:1A:2B : 3C:4D:5E
  ↑          ↑
Vendor   Device ID
```

First part → manufacturer
Second part → unique device

### Where MAC Address is Used

- Local network communication
- Used by switches
- Works at Data Link Layer (Layer 2)

### How It Works

1. You send data to another device

2. System uses:

IP → MAC (via ARP)

3. Packet delivered using MAC address

**Check MAC Address (Linux):** ip a or ifconfig

**Important Points**

- Unique for each device
- Cannot be used on internet
- Used only in local network

MAC can be **Changed (spoofed).** But original is hardware-based

## Final Summary

A MAC address is a unique hardware identifier used to identify devices in a local network and enable communication at the data link layer.

## NIC

NIC (Network Interface Card) is a hardware component that allows a computer to connect to a network. It provides the interface between the device and the network (wired or wireless).

NIC = network adapter of your computer

Without NIC: No internet / no network connection

**Examples**

- Ethernet card (LAN cable)
- Wi-Fi adapter
- Virtual NIC (in VM/cloud)

## Types of NIC

**1 Wired NIC**

Uses Ethernet cable

**Example: eth0, ens5**

**2 Wireless NIC**

Uses Wi-Fi

**Example: wlan0**

**3 Virtual NIC**

Used in VMs / cloud

**Example:**

AWS EC2 network interface

### What NIC Contains

- MAC address (unique ID)
- Hardware for sending/receiving data

**How NIC Works**

1. Receives data from network
2. Converts signals → data
3. Sends to OS

**(Reverse for sending)**

**Check NIC in Linux**

**ip a**

**Shows:**
- NIC
- MAC address
- IP address

## Key Points

Required for network connection

Works at Layer 1 & 2

Has MAC address

## Final Summary

NIC is a hardware or virtual component that connects a device to a network and enables communication using a MAC address.

## Firewall

A firewall is a security system that monitors and controls incoming and outgoing network traffic based on predefined rules.
It acts as a barrier between a trusted network and untrusted networks (like the internet).

**Simple Understanding**

**Firewall = security guard at the gate**

Allows safe traffic

Blocks harmful traffic

**Example**
**Your server:**

Port 22 (SSH) → allowed
Port 3306 (MySQL) → blocked
Only permitted traffic can enter